6.3 The Fundamental Theorem of Calculus

Name _____ Period ____ Date _____

AP Calculus 6.3 · Student sheet

I can…

- evaluate a definite integral with an antiderivative.
- find the derivative of a function defined by an integral.
- interpret the integral of a rate of change.

Vocabulary · fill in the missing word

Fundamental Theorem, Part 1 — if f is _____, $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$

Fundamental Theorem, Part 2 — $\int_a^b f(x)\,dx = F(b) - F(a)$, where F is an _____ of f

Accumulation function — a function like g(x) = _____

Part 1 · Area machine

$g(x) = \int_0^x t\,dt$.

x	1	2	3
g(x)			
$x^2/2$			

Part 2 · Notes and examples (fill in each note, then work the example)

Part 2: if F is an antiderivative of f, then $\int_a^b f(x)\,dx$ = _____.

Example 1

Evaluate $\int_1^3 (3x^2 - 2x)\,dx$.

Example 2

Evaluate $\int_0^{\frac{\pi}{2}} \cos x\,dx$.

Part 1: $\frac{d}{dx}\int_a^x f(t)\,dt$ = _____.

Example 3

$F(x) = \int_2^x \sqrt{t^2 + 9}\,dt$. Find $F'(4)$.

With the Chain Rule: $\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x)) \cdot$ _____.

Example 4

$F(x) = \int$ from 1 to x^2 of $(t^3 + 1)\,dt$. Find $F'(1)$.

Net change: the integral of a rate of change gives the net _____ in the quantity.

Example 5

Water flows in at $r(t) = 2t + 10$ L/min. The tank starts with 50 L. How much is in it at $t = 6$?

Part 3 · Partner practice

Partner 1

Evaluate $\int_0^2 (x^2 + 1)\,dx$.

Partner 2

(AP® Practice) If $F(x) = \int$ from 0 to x of $\sin t\,dt$, then $F'(\frac{\pi}{2}) =$

(A) 1 (B) 0 (C) −1 (D) $\frac{\pi}{2}$